

Simulation of Particle-Laden Thermally Buoyant Horizontal Jets

Giuliano ZENNARO* and Thomas LEPERE†

STI - Mechanical Engineering, École Polytechnique Fédérale de Lausanne (EPFL)

(Dated: August 7, 2026)

An Ansys Workbench project was configured to enable the numerical investigation of particle-laden thermally buoyant jets. The modeling approach combines a 3D steady Reynolds-averaged Navier–Stokes formulation for the carrier phase with a Lagrangian discrete phase model for particle transport. A simple test case was considered to demonstrate the capabilities of the simulation setup. The results show that, for the selected parameters, thermal buoyancy significantly influences the jet development, while its influence on the resulting sedimentation pattern is limited for the majority of particles, affecting only a small fraction. This setup provides a flexible basis for future parametric studies of particle-laden buoyant jets in environmental and industrial applications.

Keywords: Buoyant Jet, Particle Transport, Environmental Fluid Mechanics, CFD, RANS, DPM

NOMENCLATURE

$(\cdot)_\infty$	Ambient (co-flow) value	—
$(\cdot)_T$	Thermal value	—
$(\cdot)_U$	Velocity value	—
α_p	Particle volume fraction	—
$\bar{(\cdot)}$	Mean of a value	—
δ	Shear-layer thickness	m
$\Delta(\cdot)$	Difference between jet and ambient values	—
ρ_p	Particle density	kg m^{-3}
$\sigma_{(\cdot)}$	Standard deviation	—
C	Particle concentration	kg m^{-3}
d_p	Particle diameter	m
N	Number of mesh elements	—
r	Radial distance from jet centerline	m
R_{jet}	Jet radius	m
S_G	Global mesh size	m
T	Temperature	K
U	Velocity	m s^{-1}

ACRONYMS

DPM: Discrete Phase Model

DRW: Discrete Random Walk

FVM: Finite Volume Method

RANS: Reynolds-averaged Navier-Stokes

I. INTRODUCTION

Particle-laden jets are commonly found in the environment and arise from both natural and anthropogenic sources, including estuarine flushing, coastal groundwater discharge, hydrothermal vents, volcanic eruptions, wastewater outfalls, power plant discharges, and vehicle exhausts [1–3]. Their dynamics are inherently complex, as they depend on the interplay between jet momentum, buoyancy induced by density variations, particle size and density, and jet orientation relative to gravity [1, 4].

Such flows are of particular interest in environmental applications as they transport crucial particles such as pollutants, nutrients, and fine sediments [1]. A detailed understanding of particle-laden buoyant jets is therefore essential to quantify sedimentation patterns, turbulent mixing, and entrainment processes [2, 5]. Applications range from dredging operations to environmental impact assessment in contexts such as wastewater disposal and deep sea mining [2, 5].

The aim of this study is to prepare a numerical model which can be used to gain understanding in the sedimentation process under the effect of the buoyant plume. To do so, a 3D Reynolds-averaged Navier-Stokes (RANS) numerical simulation was done with Ansys Fluent to calculate the evolution of the jet under varying conditions and to compute the resulting sediment deposition maps.

II. FLOW CONDITIONS

A. Computational Domain

The flow was modeled in a rectangular prism of dimensions 100 m in length and 10 m by 10 m in cross section. A Cartesian coordinate system was adopted, with the x -axis aligned with the streamwise direction, the y -axis pointing vertically upward, and the z -axis oriented in the cross-flow direction. The jet center was located at the origin.

* giuliano.zennaro@epfl.ch; SCIPER: 344437

† thomas.lepere@epfl.ch; SCIPER: 369279

B. Boundary Conditions

All boundaries, except for the inlet and outlet, were treated as slip walls, as the primary interest is to study a free jet without wall interference. The outlet was modeled as a pressure outlet with a gauge pressure of zero.

To ensure smooth spatial gradients, the inlet velocity $U(r)$ and temperature $T(r)$ profiles were set by a hyperbolic tangent function of the radial coordinate r , representing a jet of radius R_{jet} (Fig. 1):

$$U(r) = U_{\infty} + \frac{\Delta U}{2} \left(1 - \tanh \left(\frac{r - R_{\text{jet}}}{\delta_U} \right) \right) \quad (1)$$

$$T(r) = T_{\infty} + \frac{\Delta T}{2} \left(1 - \tanh \left(\frac{r - R_{\text{jet}}}{\delta_T} \right) \right) \quad (2)$$

where U_{∞} represents the ambient co-flow velocity, used to model either sea currents or river flow at an ambient temperature T_{∞} . ΔU and ΔT denote the velocity and temperature excess of the jet relative to the ambient fluid. The ratio of the thermal boundary layer thickness δ_T to the velocity boundary layer thickness δ_U was set to unity in our case under the assumption of a fully turbulent flow. The relevant parameters are summarized in Table I. At the inlet, the turbulence intensity was set to 3%, and the turbulence length scale was specified using a hydraulic diameter equal to $2R_{\text{jet}}$.

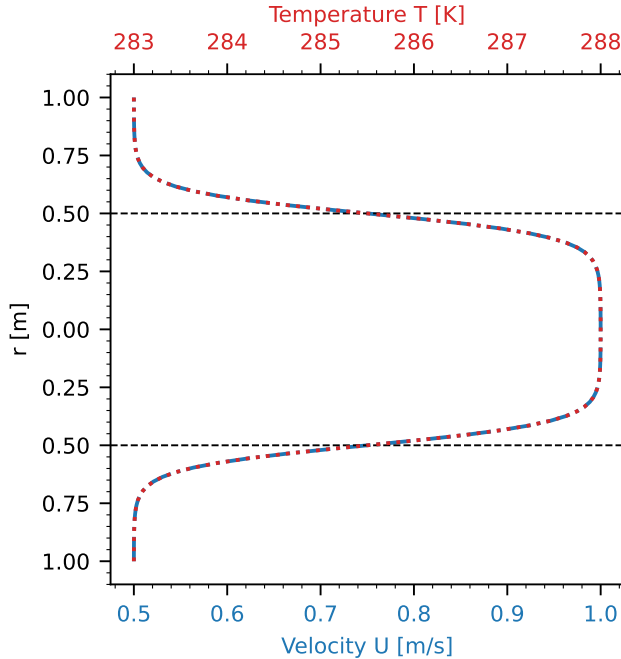


FIG. 1: Inlet profiles of axial velocity (blue dash-dotted line) and temperature (red dotted line) as functions of the radial coordinate. Profiles are prescribed using hyperbolic tangent functions, with dashed black lines indicating the jet radius R_{jet} .

C. Material Properties

The carrier phase is liquid water. Thermal buoyancy effects were accounted for using the Boussinesq approximation with a reference temperature of 283 K. The reference density was set to 999.7 kg m^{-3} , and the thermal expansion coefficient to $8.8 \times 10^{-5} \text{ K}^{-1}$.

The dispersed phase consisted of spherical particles with a material density ρ_p and a diameter d_p , injected at an inlet volume fraction of α_p (Tab. I). Particle transport was modeled using a Lagrangian approach. Parcels were injected at the inlet using a surface injection, with a total of 5×10^4 parcels uniformly distributed over a circular disk of radius R_{jet} . The parcels were initialized with the local jet velocity and temperature. Additionally, the bottom boundary was modeled as a depositional surface (trap) representing sediment settling.

III. STUDIED PARAMETERS

The study focuses on the impact of the temperature difference between the jet and the far-field flow on the particles diffusion in the fluid. The ΔT parameter was varied, taking the values $\{1, 5, 10\} \text{ K}$ (Tab. I). All other parameters were chosen arbitrarily, but could for example represent a discharge of water containing tire and road wear particles into a slow river [6].

IV. NUMERICAL METHOD

A. Type of Simulation and Solver

A 3D RANS study was conducted with Ansys Fluent, employing the Finite Volume Method (FVM) with linear elements for spatial discretization. The simulations were steady, incompressible and pressure-based, with the energy equation activated to account for thermal buoyancy effects. Turbulence was modeled using the realizable $k-\epsilon$ model. The governing equations are coupled using a least-squares cell-based method for gradient reconstruction. Second-order discretization schemes were employed for pressure, momentum, and energy, while a first order upwind scheme was used for turbulent kinetic energy and turbulent dissipation.

The particles were modeled using the Discrete Phase Model (DPM), where each computational parcel represents a collection of physical particles with identical properties. Two-way coupling was enabled to account for momentum and energy exchange between the carrier phase and the parcels. Additionally the thermophoretic force, Saffman lift force, virtual mass force, and the pressure-gradient force were taken into account. Turbulent dispersion of parcels was modeled using the Discrete Random Walk (DRW) approach with a random eddy lifetime formulation. For each injection, 20 stochastic tries were used, and the DRW time-scale constant was set to 0.15.

TABLE I: Simulation parameters used for each study case varying the parameter ΔT .

Case	U_∞ (m s^{-1})	T_∞ (K)	ΔU (m s^{-1})	ΔT (K)	R_{jet} (m)	δ_U (m)	δ_T (m)	ρ_p (kg m^{-3})	α_p (-)	d_p (m)
Case 1	0.5	283	0.5	1	0.5	0.1	0.1	1725	1×10^{-3}	7.5×10^{-4}
Case 2	0.5	283	0.5	5	0.5	0.1	0.1	1725	1×10^{-3}	7.5×10^{-4}
Case 3	0.5	283	0.5	10	0.5	0.1	0.1	1725	1×10^{-3}	7.5×10^{-4}

B. Solution Options and Computed Quantities

The velocity, temperature, and particle concentration fields were computed throughout the computational domain. Although the flow is inherently three-dimensional, the analysis focuses on selected 2D planes, namely a vertical slice corresponding to the domain mid-plane ($z = 0$) and the bottom boundary.

Each simulation was considered converged when the maximum temperature T_{max} , the mean particle concentration $\bar{C}$, and the standard deviations of temperature and particle concentration, σ_T and σ_C , respectively, varied by less than 5%. Temperature statistics were evaluated along a vertical line located on the jet centerline at $x = 25$ m, while particle statistics were computed along the bottom boundary on the mid-plane ($z = 0$) over the interval $15 \text{ m} \leq x \leq 55 \text{ m}$, corresponding to the region of predominant particle deposition.

V. MESH CONVERGENCE STUDY

A. Convergence Criterion

The mesh convergence analysis was conducted based on the same parameters defined in the previous section (T_{max} , σ_T , $\bar{C}$, σ_C). The solution obtained on the finest mesh was taken as the reference, and mesh independence was assumed when the relative deviation of all monitored quantities remained below 5%.

B. Mesh Generation

1. Global Mesh Parameters

The domain was discretized using a structured hexahedral mesh generated with the sweep method, using the inlet as the source face. To obtain a finer grid around the jet, biases were used along each edge. In the vertical and cross-flow direction, a bias factor of 2.5 was chosen, while a bias factor of 10 was applied in the streamwise direction. A size factor of global size S_G was used on each edge such that the smallest element, corresponding to the one at the inlet in the center of the jet, was approximately cubic with side length S_G .

2. Local Adaptations

To properly model the jet, previous studies suggest that a minimum of 5 mesh elements are required across the jet radius [3]. To achieve such a refinement while reducing the computational cost, a local mesh adaptation was performed on a cylindrical region twice the radius of the jet and 20 m long in the spanwise direction, representing the region of highest temperature and velocity gradients. A single level of mesh adaptation was sufficient to obtain between five and ten elements across the jet radius for the present set of parameters.

C. Mesh Convergence

The mesh convergence was assessed using the parameters of case study 2 (Tab. I). The element size S_G was varied such that the total number of elements in the domain approximately doubles between successive mesh levels. To evaluate mesh convergence, the relative deviation of several quantities (section V A) with respect to the finest mesh solution were computed and compared (Fig. 2).

The particle related quantities $\bar{C}$ and σ_C varied the most, thus their convergence behavior was further examined in terms of absolute values (Fig. 3).

D. Final Mesh

Based on the previously defined 5% criterion (section V A) and the results presented in Figures 2 and 3, a global mesh size S_G of 0.175 m was selected. With the local mesh adaptation, the jet is resolved by 11 elements along its diameter. The resulting mesh is presented in Figure 4.

Due to the structured nature of the mesh, both the skewness and orthogonal quality are near perfect, with averages of 1×10^{-4} and 1 respectively. The aspect ratio is smallest near the origin and increases to a maximum of 10 on the outlet with an average of 2.8. This was deemed acceptable as the more important regions have high quality.

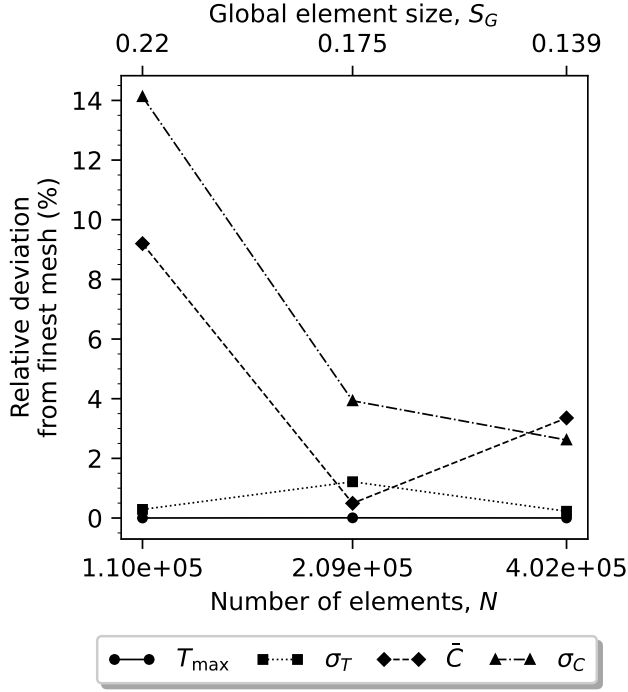


FIG. 2: Relative deviation of the monitored quantities with respect to the finest mesh solution as a function of mesh refinement.

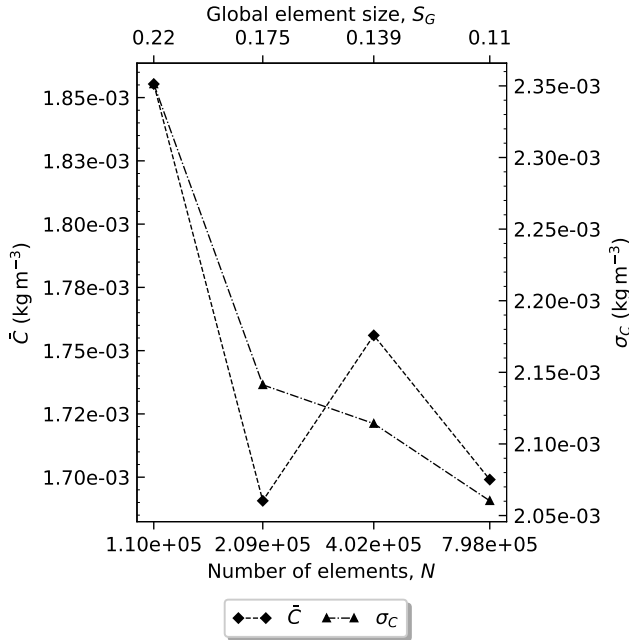


FIG. 3: Absolute values of the mean particle concentration $\bar{C}$ and concentration standard deviation σ_C as a function of mesh refinement.

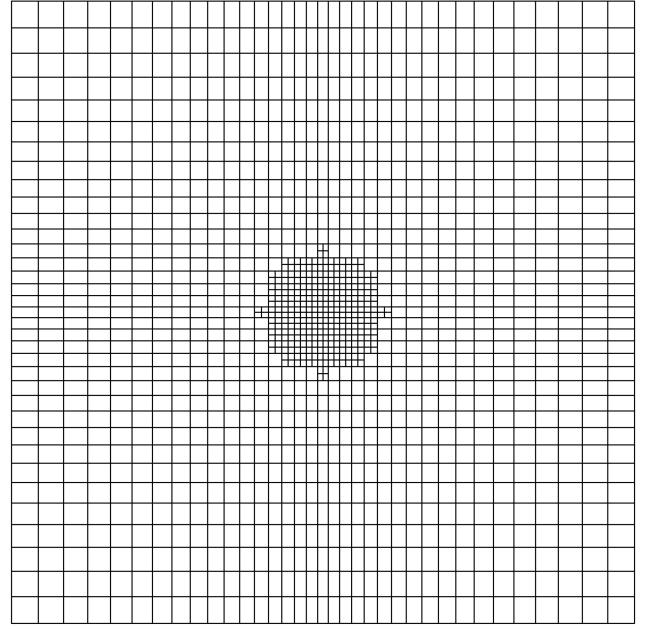


FIG. 4: Optimal mesh for the case study displaying the inlet face with the structured sweep mesh and the local adaptation around the jet.

VI. RESULTS

The trajectories along the mid-plane of the thermal and sediment plumes were computed from the fluid temperature and particle concentration fields, respectively. In both cases, the centerline was defined as the locus of the maximum field value and the plume spread as the transverse position corresponding to 20% of the local maximum for the particle plume and 80% for the temperature plume. For example for the temperature, the centerline and edge of the plume positions correspond respectively to:

$$y(x) = \arg \max_y T(x, y) \quad (3)$$

$$y(x) = \left\{ y \mid T(x, y) = 0.8 \cdot \left(\max_y T(x, y) - T_\infty \right) \right\} \quad (4)$$

The resulting trajectories are plotted in Figure 5.

While plume trajectories provide information on jet development and particle transport, they do not directly indicate the spatial extent of regions affected by particle deposition. To assess and compare potentially impacted zones, sediment deposition maps were constructed on the bottom boundary (Fig. 6).

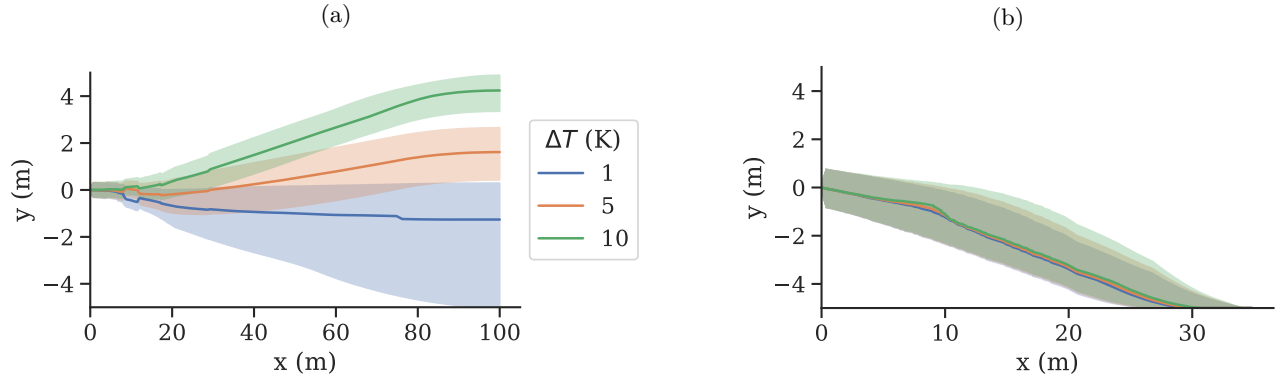


FIG. 5: Trajectories of the (a) thermal plume and (b) sediment plume centerlines, together with the corresponding plume spreads, for different jet temperature values.

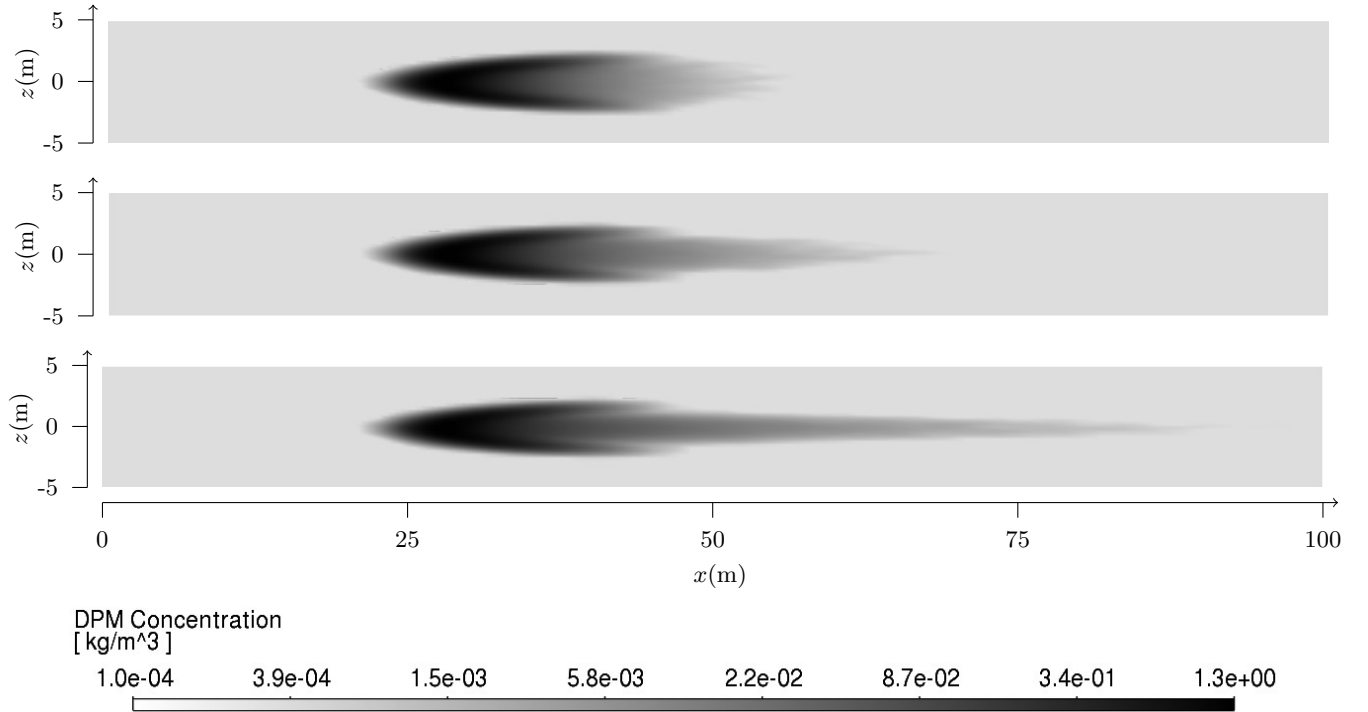


FIG. 6: Sediment deposition maps for different jet temperatures. From top to bottom $\Delta T = \{1, 5, 10\}$ K.

VII. DISCUSSION

The results presented in the previous section indicate that thermal buoyancy strongly influences the evolution of the thermal plume, while the effect is limited for the bulk of the particles (Fig. 5). Nonetheless, the sedimentation map (Fig. 6) reveals that, although most particles settle within a similar region regardless of jet temperature, a small fraction can be entrained by the vertical motion of the thermal plume and remain suspended for over twice the distance.

The first observation should be taken with caution as to not overgeneralized it. Although thermal effects appear to be modest for most particles in our parameter range, different flow and particle regimes such as different Richardson numbers, different particle densimetric Froude numbers, or different Stokes numbers, may result in very different behavior.

Finally, the fact that some particles get entrained in the vertical motion of the plume suggests that thermal effects must be considered in applications such as environmental impact assessments. Even if only a small pro-

portion of particles are entrained, underestimating the sedimentation zone may lead to unforeseen and unaccounted contaminated regions.

VIII. CONCLUSION

A numerical simulation was performed to study particle-laden thermally buoyant horizontal jets using a coupled Eulerian–Lagrangian approach. Results from a representative test case indicate that thermal buoyancy strongly influences jet evolution, while its impact on sediment deposition remains limited for most particles under the conditions considered.

Future investigations should explore a wider range of flow and particle parameters, as well as the effects of domain confinement and inlet turbulence conditions. Future work should include validation against experimental or field data to further assess the predictive capability of the proposed simulations. Moreover, the use of a steady RANS formulation represents a simplifying assumption whose influence on plume dynamics and particle dispersion merits additional investigation.

-
- [1] Z. He, H. Zhang, H. Zhang, S. U. Okon, and R. Zhu, Particle dynamics in horizontal buoyant jets within linearly stratified environments, *Environmental Fluid Mechanics* **25**, 10.1007/s10652-025-10035-3 (2025).
 - [2] N. Mingotti and A. W. Woods, Dynamics of sediment-laden plumes in the ocean, *Flow* **2**, 10.1017/flo.2022.12 (2022).
 - [3] Y. J. Suzuki, T. Koyaguchi, M. Ogawa, and I. Hachisu, A numerical study of turbulent mixing in eruption clouds using a three-dimensional fluid dynamics model, *Journal of Geophysical Research: Solid Earth* **110**, 10.1029/2004jb003460 (2005).
 - [4] S. Qin, G. Chen, F. Shan, W. Liu, and B. Zang, Large eddy simulation of round jets with mild temperature difference, *International Journal of Mechanical Sciences* **283**, 109649 (2024).
 - [5] D. Robinson, M. Wood, M. Piggott, and G. Gorman, Cfd modelling of marine discharge mixing and dispersion, *Journal of Applied Water Engineering and Research* **4**, 152–162 (2015).
 - [6] P. M. Mayer, K. D. Moran, E. L. Miller, S. M. Brander, S. Harper, M. Garcia-Jaramillo, V. Carrasco-Navarro, K. T. Ho, R. M. Burgess, and L. M. Thornton Hampton, Where the rubber meets the road: Emerging environmental impacts of tire wear particles and their chemical cocktails, *Science of The Total Environment* **927**, 171153 (2024).